Rossini Martyr

Data Science | Analytics | Engineering

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MOTIVATION

Data Scientist and licensed Professional Engineer (P.Eng) with 5+ years of data analytics experience and 5+ years of quantitative project and stakeholder management in engineering consulting. Proven capability building end-to-end predictive machine learning solutions across financial risk, retail analytics, and insurance domains.

SKILLS & TOOLS

Languages & Databases: SQL, Python, AWS Redshift
Machine Learning: XGBoost, Gradient Boosted Trees, Random Forest, Anomaly Detection, SHAP Explainability
Statistics & Experimentation: Hypothesis (A/B) Testing, Causal Impact Analysis, Causal Inference
MLOps & Cloud: Docker, FastAPI, AWS Sagemaker, Streamlit, Git, AutoML

WORK EXPERIENCE

Senior Engineer, Fast + Epp (Toronto, Canada) February 2024 – Present

- Led cross-functional teams across multi-disciplinary $100M+ capital projects, managing project risk, and translating complex technical outputs from data analysis into actionable guidance for stakeholders
- Managed project finances by working within established internal budgets to satisfy stakeholder requirements while maintaining profitability
- Mentored and supervised junior and intermediate technical staff, enforcing peer review standards, conducting performance evaluations, and ensuring project delivery within target timelines

Senior Engineer, Arup Canada (Toronto, Canada) April 2016 – February 2024

- Built scalable SQL and Python analytical scripts leveraging AWS Redshift to ingest, process, and analyze multi-million row datasets, providing instant results from each cloud compute job.
- Automated analytical design workflows by leveraging custom Python scripts via software API calls, reducing model creation time by 30% and eliminating manual data preparation errors
- Directed and evaluated junior engineers, establishing standardized QA/QC validation frameworks for numerical modeling and providing career development guidance

RELEVANT EXPERIENCE

Quantifying Sales Uplift using Causal Impact Analysis (Github)

- To quantify the financial uplift of a new membership tier, I conducted a **Causal Impact Analysis** to measure if waiving per-order delivery fees increased overall customer spending for a grocery retailer. I compared members against a control group of non-members using three months of pre- and post-campaign transaction data. The **analysis revealed** a statistically significant (95%) **sales uplift of 41.1% among club members**, confirming that eliminating delivery friction drove higher overall revenue.

RELEVANT EXPERIENCE (CONT'D.)

Assessing Campaign Performance Using A|B Testing (Github)

- To **optimize campaign ROI** for a new membership tier, I conducted an **A/B test** to determine if expensive promotional mailers drove higher signup rates than cheaper alternatives. Utilizing a **Chi-Square Test for Independence**, the analysis revealed that while the expensive mailer yielded a higher raw signup rate, the difference was not statistically significant. This showed that the expensive marketing materials did not conclusively guarantee higher conversion rates, **saving advertising costs.**

Credit Card Transaction Fraud Detection (Github)

- To minimize financial losses from credit card fraud, I built a machine learning model using **XGBoost** to automatically classify fraudulent credit card transactions. Achieving a **Precision-Recall AUC of 0.865**, this model detects fraudulent charges **improving baseline model performance by 15%**. I containerized the model and **deployed a production-ready API using Docker and FastAPI** to enable real-time risk assessment.

Building an SQL AI Agent for Retail Client (Github)

- To **automate complex query generation** for non-technical stakeholders, I developed an **autonomous AI SQL Agent** using Python, LangChain, and LLMs connected to a live PostgreSQL database. By integrating strict validation guardrails and execution constraints to ensure data security and query accuracy, I eliminated analytical bottlenecks while **delivering instant data insights from natural language inputs.**

EDUCATION

MASc (Structural Engineering) (GPA: 3.94/4.00) 2013 - 2016
University of Toronto
- *P*ublished M.A.Sc research in prestigious structural engineering journal

BASc (Civil Engineering) (GPA: 3.88/4.00) 2008 - 2013
University of Toronto
- Graduated #1 in Civil Engineering

COURSES, CERTIFICATES AND QUALIFICATIONS

Data Science & AI Professional Certification 2026
Data Science Infinity

- **Actionable Learnings:** Advanced data visualizations using Tableau. Designing, executing, and evaluating rigorous A/B tests and causal impact analyses to measure business impact and drive data-informed product decisions. Packaging machine learning models into production-ready APIs and deploying them to AWS cloud environments using SageMaker and AutoML.

Data Science Professional Certification 2025 - 2026
IBM

- **Actionable Learnings:** Querying databases and manipulating data structures using SQL. Automated web scraping using BeautifulSoup and web APIs. Utilizing Pandas and NumPy for data wrangling, and Matplotlib, Folium, and Plotly to build interactive geospatial and statistical visualizations. Implementing regression, classification, and clustering algorithms using Scikit-learn. Navigating GitHub for version control. Leveraging GenAI tools to accelerate code generation, debugging, and data exploration.

Professional Engineer in Ontario 2018 - Current
PEO